

Traditionally Used Food Plants for Eri Silkworm

Introduction

Silk is a natural fiber obtained from the cocoons of silkworms. It is a soft, strong, and lustrous fiber that is used to manufacture various textile products, including clothing, household furnishings, and medical materials. The mulberry silkworm (*Bombyx mori*), domesticated in China approximately 5,200 years ago, is one of the major producers of silk fiber. Silkworms produce silk by secreting a liquid protein that hardens into fibers when exposed to air. Silk is valued for its versatility, durability, breathability, and lightweight nature, making it suitable for both warm and cold climates.

Sericulture, the practice of rearing silkworms and producing silk, is an environmentally friendly agro-industrial activity carried out in many regions of the world, particularly in Asia and the Pacific, which together account for the majority of global silk production. Besides its economic significance, sericulture preserves cultural traditions and provides employment opportunities.

India produces several varieties of silk, including oak tasar, muga, mulberry, eri, and tasar silk. Among the non-mulberry silk-producing species, *Antheraea mylitta* (tasar silkworm), *Antheraea pernyi* (Chinese oak silkworm), *Antheraea assamensis* (muga silkworm), and *Samia ricini* (eri silkworm) are the most important.

Endi silk is indigenous to Assam and is traditionally reared by women of the Bodo community, as well as by other tribal and non-tribal groups. The practice contributes significantly to the livelihoods of rural households and forms an essential part of their socio-cultural and economic life. The term "Endi" originates from the castor plant, locally known as "Endi," which serves as the primary food source for the silkworm. In Assamese, Endi silk is commonly known as Eri silk.

The eri silkworm, *Samia ricini*, is primarily found in Northeast India, especially in Assam, where a rich diversity of host plants supports its cultivation. The northeastern region is considered a natural hotspot for serigenous insects and their host plants.

Eri silk is a natural, hypoallergenic, and non-toxic fiber. It is breathable, keeping the body cool in summer and warm in winter. Due to its durability and versatility, it is used in a wide range of garments and household products. Eri silk is often referred to as "Ahimsa silk" because silk extraction does not require killing the pupae.

According to the Central Silk Board Annual Report (2021–22), India produced 9,085 metric tons of non-mulberry raw silk, of which Eri silk accounted for approximately 81%. Assam and the Bodoland Territorial Region (BTR) alone contributed about 5,465 metric tons. The industry creates employment opportunities in host plant cultivation, silkworm rearing, reeling, spinning, weaving, and the marketing of silk and its by-products.

Overview

The settlements of Raimona, Bongaon, Dotma, Anthihara, Naigaon, and Harinaguri documented six traditional food plants used for rearing Eri silkworms (*Samia ricini*). Among these plants, castor (*Ricinus communis*) is the principal host plant and serves as the primary source of food for the silkworm. The remaining plants are used as secondary food sources when castor leaves are unavailable.

The six food plants identified are:

1. Castor
2. Kesseru
3. Tapioca
4. White Teak (Gamari)
5. Cluster Fig
6. Cabbage

Castor leaves are preferred because they produce high-quality cocoons. However, since castor leaves may not be available throughout the year, silkworm rearers use alternative host plants to ensure continuous rearing.

Different villages use different combinations of host plants:

- Raimona: Castor, Tapioca, and White Teak.
- Bongaon: Castor and White Teak.
- Anthihara: Castor, White Teak, and Cluster Fig.
- Naigaon and Harinaguri: A larger variety of food plants due to their proximity to Bodoland Silk Park.
- Dotma: Five different food plants, including cabbage during periods of castor shortage.

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Sl. No.	Name of Food Plant	Scientific Name	Family	Local Name (Bodo)
1	Castor	<i>Ricinus communis</i> L.	Euphorbiaceae	Endi
2	Kesseru	<i>Heteropanax fragrans</i> (Roxb.) Seem.	Araliaceae	Karong

3	Tapioca	<i>Manihot esculenta</i> Crantz	Euphorbiaceae	Tasumbli
4	White Teak	<i>Gmelina arborea</i> Roxb.	Corniaceae	Gambhari
5	Cluster Fig	<i>Ficus racemosa</i> L.	Euphorbiaceae	Dhumbri
6	Cabbage	<i>Brassica oleracea</i>	Brassicaceae	Kobi